Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science &Engineering

B. E.(COE/CSE) : EST	**Course Code: UCS510 / ULC303** **Course Name: Computer Architecture and Organization**
17/Dec/2025	**9:00 AM – 12:00PM**
Time: 3 Hours, M. Marks: 42	**Faculty: SHI/AAS/YAS/ADL/JSK**

Note: All questions carry equal marks. Attempt any 3 out of 4 questions. Attempt questions in sequence, attempt all subparts of question at same place and assume for any missing data.

Q. No	Questions	Marks
Q1.(a)	Instruction execution in a processor is divided into a multi-stage pipeline: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Execute (EX), and Write Back (WB). A. Each stage takes the following times: IF = 10 ns, ID = 10 ns, OF = 12 ns, EX = 30 ns, WB = 8 ns. Assume there is no delay in buffering between consecutive stages. Further the pipeline is modified by dividing the EX-stage into two stages (EX1 and EX2), with each stage taking 15 ns. a. Determine the cycle time for both first and second pipeline b. Calculate the speedup achieved by both pipeline for executing 100 independent instructions without any hazards. B. A two-word instruction is stored in memory starting at address W = 500. The second word of the instruction at address W + 1 = 501 contains the address field Y = 200. The operand is stored at address Z. An index register contains the value X = 40. Determine the effective address (Z) of the operand for the following addressing modes: (i) Direct addressing (ii) Indirect addressing (Given that memory location 200 contains value 900) (iii) Relative addressing	**4+3** **CO2** **L5**
Q1.(b)	A. Consider the following instruction sequence executed on a 5-stage RISC pipeline (IF → ID → EX → MEM → WB) with forwarding support but no branch prediction: I1: LW R1, 0(R2) I2: ADD R3, R1, R4 I3: SUB R5, R3, R6 I4: BEQ R5, R0, LABEL I5: AND R7, R1, R8 a. Identify **all the pipeline hazards** present in the given instruction sequence. b. Suggest an appropriate **solution** for each identified hazard. B. State four logical differences between RISC and CISC. Support your answer with suitable example.	**5+2** **CO2** **L6**
Q2.(a)	A computer system is equipped with a L1 data cache and a main memory, and you are interested in understanding the importance of the principle of locality in justifying the use of cache memory. In this analysis, consider only data memory accesses. The access latencies (measured in CPU cycles) are given as follows: 1. Cache hit: 1 cycle 2. Cache miss (with cache enabled): 105 cycles 3. Main memory access with cache disabled: 100 cycles Answer the following questions: A. If a program is executed with an overall cache miss rate of 5%, calculate the average memory access time (AMAT) in CPU cycles. B. Now consider a program that is deliberately designed to generate completely random data accesses, exhibiting no spatial or temporal locality. To achieve this, the program repeatedly accesses random elements of an array of size 256 MB, which entirely resides in main memory. The array indices are generated using a uniform random number generator. If the L1 data cache size is 64 KB, determine the average memory access time (in CPU cycles) for this program. Clearly justify your reasoning.	**3+4** **CO3** **L5**

Q2.(b)	Consider a memory system that uses a 32-bit byte-addressable address space and a cache with a block size of 64 bytes. A. Assume a direct-mapped cache with a tag field of 20 bits. Show the address format and determine: a) Number of blocks in main memory b) Number of cache blocks c) Size of extra memory required to store cache tags B. Assume a four-way set-associative cache with a tag field of 9 bits. Show the address format and determine: a) Number of sets in cache b) Number of blocks per set c) Total number of cache blocks d) Size of extra memory required to store cache tags	3+4 CO3 L6
Q3.(a)	Transferring data between input/output (I/O) devices and memory can slow performance if the CPU manages every step. To overcome this, a Direct Memory Access (DMA) Controller is used. A. Explain the step by step working of DMA when memory needs to transfer files of size 70MB to I/O. Explain each involved signal, registers and pins by giving suitable diagram. B. Compare different modes of transfer in DMA with respect to: • CPU starvation • Memory latency • Suitability for real-time applications **(Discuss at least 2 modes)**	5+3 CO4 L5
Q3.(b)	A. Differentiate between Interrupt Driven (I/O) and Programmed I/O. **(Give any 6 differences)** B. DMA Module is transferring characters to memory using cycle stealing, from a device transmitting at 9600 bps. The processor is fetching instructions at the rate of 1 million instructions per second (1 MIPS). By how much will the processor be slowed down due to DMA activity?	3+3 CO4 L5
Q4.(a)	A Datapath contains six 16-bit registers: R0, R1, R2, R3, R4, and R5. The system must support the following register transfers into R5, controlled by five mutually-exclusive timing/control variables TT4: T0: R5 ← R0 T1: R5 ← R1 T2: R5 ← R2 + R3 (Ignore carry) T3: R5 ← (R4 ∧ R1) T4: R5 ← MUX-A(R0, R1, R2, R3), with S1 and S0 being 2-bit select lines generated from a control decoder. All timing variables (T0 – T4) are mutually exclusive. R5 has a load-enable input (LD5) that must be controlled correctly. Only one operation should reach R5 during any timing interval. The hardware must be designed so that the final multiplexer selecting R5's input must be a 6-to-1 mux, even though only 5 timing variables exist. The last input is none active. Answer the following questions: A. Construct the truth table that maps timing variables T0 to T4 to the final 6-to-1 multiplexer's select inputs (M2 M1 M0). B. Design the combinational logic equations for: the final mux select bits (M2, M1, M0), and the load-enable signal LD5 using minterms.	5+4 CO2 L6
Q4.(b)	Consider the question given in 4a, Draw the complete Datapath block diagram that implements all five transfers into R5, the final 6-to-1 mux feeding R5, proper lines from timing variables T0–T4, and load-enable logic for R5	5 CO2 L6